

Take-Home Technical Assessment

Senior AI Software Architect / Agentic AI Engineer

Objective: Build a small but production-minded Agentic AI MVP demonstrating software architecture, scalable backend design, and effective use of an AI coding agent.

Scenario

Create a backend service for a Real Estate AI Assistant that can (1) answer FAQs, (2) search properties, and (3) schedule property viewings using AI tool/function calling.

Deliverables

- GitHub repository with commit history.
- Architecture diagram (1 page max).
- Technical description of the proposed software architecture
- README with setup instructions and design decisions.
- Working REST API.
- Basic admin page (like user management, prompt, etc.) for the app
- Short (5–10 min) demo video.
- Evidence of AI coding-agent usage (prompts, generated code, and your review/corrections/markdown).

Minimum Requirements

- The AI must autonomously choose the appropriate tool (no keyword-based if/else routing).
- Implement at least three tools: SearchProperty, BookViewing, EscalateToHuman.
- Design the backend to be easily extensible for future agents and services.
- Explain how your architecture would scale to 100× traffic, in a video, less than 5 minutes.
- Include basic logging and error handling.
- All in Python

Evaluation Criteria

Area	Weight
Software Architecture	30%
Agentic AI Design	25%
Backend Quality	20%
AI Coding-Agent Workflow	15%
Documentation & Communication	10%

Notes: You may use any language, framework, LLM provider, or orchestration library. We value design decisions, trade-off explanations, maintainability, and engineering judgment more than feature completeness.